

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Quality Control Overview and Interpretation

Steps for Exploring Results

- QC metrics
- Sample Report
 - Metrics
 - Analysis
- Accounting for background

CZ ID Help Center: [Analyzing Nanopore Data](#)

Outline

- QC metric definitions
- How QC metrics help inform wet lab experiments and data interpretation
- Perform QC within CZ ID (**look at your data!**)

Quality Control

- Were there enough reads / bases?
- Did my samples pass QC filtering? (basecall quality, length, complexity)
- How many reads / bases were removed during data preprocessing? (e.g., host filtering)
- How many reads / bases were used for microbe identification?

CZ ID QC Metrics

Total Reads: Total reads uploaded to the pipeline.

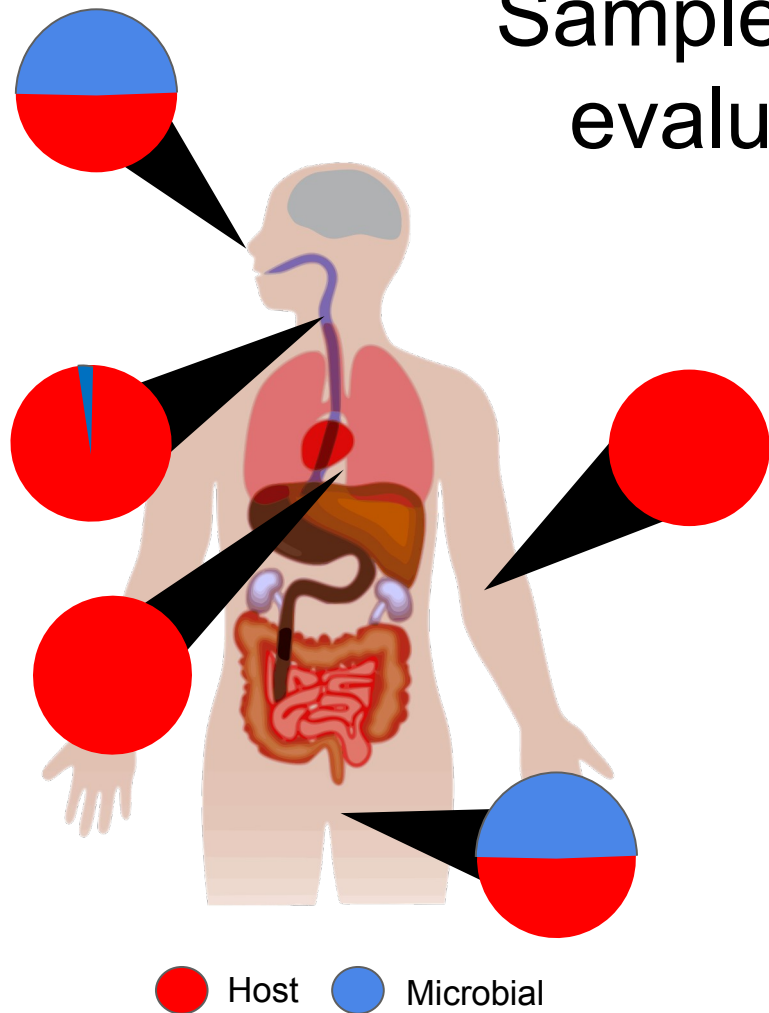
Passed QC: Percentage of reads remaining after QC filtering to remove low quality bases (phred score < 9), short reads (< 100 bp), and low complexity reads.

Subsampled Fraction: After QC filtering and host/human data removal, the remaining reads are subsampled to 1 million.

Passed Filters: Reads used for analysis. Percentage of reads that passed QC, host/human filtering, and subsampling compared to the initial reads input.

Note: The Project Page only provides stats at the read level. If you are interested in the base pair level, see the Sample Overview report available through [bulk download](#) or look at the pipeline details on the Sample Report page.

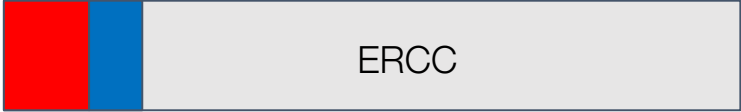
Sample type matters when evaluating QC metrics



Sample Type	Host Signal	Microbial Signal
CSF	> 99%	<< 5% with infection
Tracheal Aspirate	95%	5%
Nasal or oral	50%	50%
Whole blood	> 99%	<< 1% (with infection)
Rectal or Vaginal	50%	50%

*Concepts introduced in detail in previous modules
(Introduction to mNGS; mNGS Study Design)

QC metrics affected by sample type

Sample Type	Typical Distribution 	QC Trend
Stool, NP Swab, Environmental Samples, Sputum, Urine (Higher diversity & DNA/cDNA input)		• Higher microbial reads • Lower DCR
Serum, CSF, Plasma, Negative Controls (Lower diversity & DNA/cDNA input)		• Lower microbial reads • Higher DCR (PCR artifact) • Higher background contamination

**ERCC evaluation is not available for ONT in CZ ID, however the concepts apply*

QC metrics not affected by sample type

- **% Passed QC (fastp)**

- Should not inherently vary by tissue type
 - Per base quality (Q score)
 - Adapter dimer
 - Length
 - Complexity
- May require knowledge of sample processing for interpretation

- **Total reads**

- Not affected by sample type
- Dependent on input DNA/cDNA and sequencing depth

Project Page QC Metrics

ASTMH_Demos

Private project ⓘ 4 members [Share](#) [Add Data](#)

0

Search My Data...

Q

[Samples](#) 14

ⓘ

Taxon Filters

TAXON ▾

ANNOTATION ▾

Metadata Filters

LOCATION ▾

TIMEFRAME ▾

VISIBILITY ▾

HOST ▾

SAMPLE TYPE ▾

Metagenomics 5 [Metagenomics - Nanopore](#) 4 Consensus Genomes 4 Antimicrobial Resistance 5

0 Selected |

4 samples

☐	Sample	Created On ▾	Host	Location	Passed Filters	Passed QC	Total Reads	ⓘ
☐	SRR22762466 COMPLETE karyna Rosario Cora ASTMH_Demos	2024-08-23 2 months ago	Human	San Francisco Bay	25,034 98.18%	98.4%	25,497	
☐	44-Idseq-mosq_397581_reads_nh COMPLETE karyna Rosario Cora ASTMH_Demos	2023-10-04 a year ago	Mosquito	California	813,375 100%	100%	813,375	
☐	28A-idseq-mosq_397583_reads_nh COMPLETE karyna Rosario Cora ASTMH_Demos	2023-10-03 a year ago	Mosquito	California	204,968 100%	100%	204,968	

DESCRIPTION

EDIT

⌵

Datasets to showcase CZ ID at ASTMH 2023

OVERALL

⌵

Samples

14

Projects

1

Avg. reads per sample

557,404

Avg. reads passing filters per sample

553,703

DATE CREATED

⌵

2023-09-14

2024-09-21

Sample-level QC

< RR177 ONT 150fmol load - 2-24h

Nanopore mNGS Pipeline

1_XPR1FFSSV

Nanopore

Taxon name

Name Type: Scientific

Categories

Threshold filters

Reads

Pathogen Flags

859 rows passing the above filters, out of 940 total rows

>	Taxon	bPM	b	r	contigs
>	Staphylococcus (59 bacterial species: 7)	9,342.4 0.0	4,394,831 0	8,563 0	
>	Corynebacterium (41 bacterial species: 1)	3,254.8 2,463.8	1,531,116 1,159,025	2,791 2,018	
>	Respirovirus (1 viral species: 1)	522.6 294.6	245,821 138,578	324 188	
>	Cutibacterium (2 bacterial species)	216.4 0.0	101,786 0	209 0	
>	Malassezia (6 eukaryotic species)	173.3 0.0	81,513 0	227 0	
>	Haemophilus (6 bacterial species: 2)	104.2 0.0	49,036 0	90 0	
>	Massilia (14 bacterial species)	82.0 0.0	38,561 0	50 0	
>	Streptococcus (13 bacterial species: 4)	67.0 0.0	31,523 0	102 0	
>	Pseudomonas (35 bacterial species: 2)	66.0 0.0	31,029 0	61 0	
		47.8	22,483	43	

1_XPR1FFSSV

Metadata

Pipelines

Notes

PIPELINE INFO

Analysis Type

Sequencing Platform

Pipeline Version

Guppy Basecaller Version

NCBI Index Date

Host Subtracted

Total Reads

ERCC Reads

Passed Filters

Unmapped Reads

Passed Quality Control

Date Processed

BASES REMAINING

ERCC SPIKE-IN COUNTS

DOWNLOADS

Metagenomic

Nanopore

v0.7

View Pipeline Visualization

hac

2024-02-06

Human

1,070,772

--

22,973 (2.15%)

5,138

97.30%

2025-02-03

Pipeline Table

Pipeline Step	Function	If Many Reads Lost During this Step...
fastp	Low quality read removal Low complexity read removal Note: adapter trimming is not completed in CZID. It must be done in ONT MinKNOW/Dorado, prior to upload to CZID	Many reads of low-quality (Phred scores, GC content); check sequencing run - suggest reviewing sequencing run report Reads are relatively low-complexity, containing homopolymer nucleotides or simple sequence repeats Many short reads: possible degradation, low quality RNA.
minimap2	Host and human filtering	Large proportion of reads mapped to the host/human reference genome; this will vary by sample type and sample pre-processing
seqtk	Subsampling to reduce computational burden	Sample received a lot of reads during sequencing and/or has a high proportion of reads that are microbial

Sample-level QC

< RR177 ONT 150fmol load - 2-24h Nanopore mNGS Pipeline

1_XPR1FFSSV ▾

Nanopore

Taxon name Name Type: Scientific ▾ Categories ▾ Threshold filters ▾ Read length ▾

Pathogen Flags ▾

859 rows passing the above filters, out of 940 total rows

>	Taxon	bPM ▾	b	r	contig
>	▷ Staphylococcus (59 bacterial species: ● 7)	9,342.4 0.0	4,394,831 0	8,563 0	
>	▷ Corynebacterium (41 bacterial species: ● 1)	3,254.8 2,463.8	1,531,116 1,159,025	2,791 2,018	
>	▷ Respirivirus (1 viral species: ● 1)	522.6 294.6	245,821 138,578	324 188	
>	▷ Cutibacterium (2 bacterial species)	216.4 0.0	101,786 0	209 0	
>	▷ Malassezia (6 eukaryotic species)	173.3 0.0	81,513 0	227 0	
>	▷ Haemophilus (6 bacterial species: ● 2)	104.2 0.0	49,036 0	90 0	
>	▷ Massilia (14 bacterial species)	82.0 0.0	38,561 0	50 0	
>	▷ Streptococcus (13 bacterial species: ● 4)	67.0 0.0	31,523 0	102 0	
>	▷ Pseudomonas (35 bacterial species: ● 2)	66.0 0.0	31,029 0	61 0	
		47.8	22,483	43	

1_XPR1FFSSV

Metadata Pipelines Notes

PIPELINE INFO

Analysis Type Metagenomic

Sequencing Platform Nanopore

Pipeline Version v0.7 [View Pipeline Visualization](#)

Guppy Basecaller Version hac

NCBI Index Date 2024-02-06

Host Subtracted Human

Total Reads 1,070,772

ERCC Reads --

Passed Filters 22,973 (2.15%)

Unmapped Reads 5,138

Passed Quality Control 97.30%

Date Processed 2025-02-03

BASES REMAINING

ERCC SPIKE-IN COUNTS

DOWNLOADS

BASES REMAINING

Filtering Step	Bases Remaining	% Bases Remaining
Validate Input	470,419,960	100.00%
Quality Filter	468,702,545	99.63%
Host Filter	10,101,022	2.15%
Human Filter	10,101,022	2.15%
Subsampling	10,101,022	2.15%

Go to worksheet:
Quality control
step 1 to 5

Current Worksheet: CZ ID Training Reference Tables & Worksheets_ONT, pg 5-9

Next Slides: DL6-O_Sample Report Metrics for ONT